

DSA0601-Visualization Matching Game – Detailed Answer

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Note: I matched the charts based on the actual figures shown in the uploaded image. The correct answers are different from the previous sheet because the chart order and descriptions are different here.

Introduction

Data visualization is the process of representing data using graphical forms such as bar charts, pie charts, scatter plots, heatmaps, box plots, histograms, line charts, bubble charts, stacked bar charts, and area charts. Each type of visualization is designed to communicate a particular type of information.

In this activity, ten different charts labeled **A to J** are given on the left side, while ten descriptions numbered **1 to 10** are given on the right side. The objective is to identify the type of each chart and match it with the description that best explains its purpose.

A. Pie Chart

Matching Answer: A → 3

Description 3:

“Shows how parts contribute to a whole.”

Explanation

Chart A is a **Pie Chart**. The chart is circular and divided into several sectors or slices. Each slice represents a part of the total data.

The complete circle represents **100% of the data**, while each individual section represents a proportion of that total.

For example, if a company's total budget is divided among different departments, a pie chart can show what percentage of the total budget is spent by each department.

Characteristics of a Pie Chart

- It is circular in shape.
- It is divided into sectors.
- The complete circle represents the whole.
- Each sector represents a part of the whole.
- Larger sectors represent larger proportions.
- It is commonly used for percentage data.

Examples

A pie chart can be used to represent:

- Monthly expenses.
- Market share.
- Population percentage.
- Student distribution.
- Sources of energy.
- Company budget allocation.

Conclusion

Since Chart A represents different sections of one complete circle, it clearly shows how different parts contribute to a whole.

Therefore: $A \rightarrow 3$

B. Scatter Plot

Matching Answer: B → 6

Description 6:

“Shows the relationship between two continuous variables.”

Explanation

Chart B is a **Scatter Plot**. It consists of many individual points positioned according to values on the X-axis and Y-axis.

A scatter plot is mainly used to study the relationship or correlation between two numerical variables.

For example, we can use a scatter plot to study the relationship between:

- Study hours and examination marks.
- Height and weight.
- Advertising cost and sales.
- Temperature and electricity consumption.
- Experience and salary.

Characteristics of a Scatter Plot

- Uses individual points.
- Has an X-axis and Y-axis.
- Represents two numerical variables.
- Helps identify relationships.
- Can show positive or negative correlation.
- Can identify clusters and unusual observations.

Interpretation

If the points generally move upward from left to right, there may be a **positive relationship**.

If the points generally move downward, there may be a **negative relationship**.

If the points are randomly scattered, there may be little or no relationship.

Conclusion

Chart B contains individual points representing two continuous variables. Therefore, it matches Description 6.

Therefore: B $\rightarrow$ 6

C. Heatmap

Matching Answer: C $\rightarrow$ 10

Description 10:

“Shows magnitude across two categories using color intensity.”

Explanation

Chart C is a **Heatmap**. A heatmap represents numerical values using different colors or different intensities of color.

The rows and columns represent two categories, while the color of each cell represents the magnitude of the value.

For example, a heatmap can show student performance across:

- Different subjects.
- Different semesters.

The darker cells may represent higher scores, while lighter cells may represent lower scores.

Characteristics of a Heatmap

- Uses a rectangular grid.
- Contains rows and columns.
- Uses color to represent values.
- Darker or stronger colors can indicate higher values.
- Lighter colors can indicate lower values.
- Useful for identifying patterns quickly.

Examples

Heatmaps can be used for:

- Sales by product and month.
- Temperature by location and time.
- Student marks by subject and semester.
- Website activity by day and hour.
- Correlation matrices.
- Attendance patterns.

Conclusion

The grid-like structure and varying color intensity in Chart C identify it as a heatmap.

Therefore: C $\rightarrow$ 10

D. Bar Chart

Matching Answer: D $\rightarrow$ 2

Description 2:

“Compares amounts across separate categories.”

Explanation

Chart D is a **Bar Chart**. A bar chart uses separate rectangular bars to represent different categories.

The height or length of each bar represents the numerical value associated with that category.

For example, a company can use a bar chart to compare sales of different products.

Characteristics of a Bar Chart

- Represents categorical data.
- Uses separate bars.
- Each bar represents one category.
- Bar height represents the numerical value.
- Makes comparison between categories easy.

Examples

Bar charts can be used to compare:

- Sales of different products.
- Marks of students.
- Population of different cities.
- Number of employees in departments.
- Monthly revenue.
- Number of customers.

Advantages

The main advantage of a bar chart is that differences between categories can be identified quickly.

For example, if one bar is much taller than the others, that category has a larger value.

Conclusion

Chart D contains separate vertical bars representing different categories. Therefore, it matches Description 2.

Therefore: D $\rightarrow$ 2

E. Bubble Chart

Matching Answer: E $\rightarrow$ 9

Description 9:

“Shows three variables at once using position and size.”

Explanation

Chart E is a **Bubble Chart**. A bubble chart is similar to a scatter plot, but it adds a third variable through the size of the bubbles.

Three variables can be represented:

1. X-axis position represents the first variable.

2. Y-axis position represents the second variable.
3. Bubble size represents the third variable.

Example

A bubble chart can represent:

- X-axis → Sales.
- Y-axis → Profit.
- Bubble size → Number of customers.

Therefore, one graph can communicate three different numerical variables.

Characteristics

- Uses circular points or bubbles.
- Position represents two variables.
- Bubble size represents another variable.
- Useful for comparing several observations.
- Can display more information than a simple scatter plot.

Applications

Bubble charts can be used to compare:

- Population, income, and area.
- Sales, profit, and customer count.
- GDP, population, and life expectancy.
- Product price, sales, and market size.

Conclusion

The different bubble sizes in Chart E indicate that size is being used to represent an additional variable.

Therefore: E → 9

F. Box Plot

Matching Answer: F → 4

Description 4:

“Compares distributions across groups and highlights outliers.”

Explanation

Chart F is a **Box Plot**, also known as a **Box-and-Whisker Plot**.

A box plot is used to summarize the distribution of numerical data. It is particularly useful when comparing several groups.

A box plot can show:

- Minimum value.
- First quartile.
- Median.
- Third quartile.
- Maximum value.
- Potential outliers.

Important Components

The rectangular box represents the middle portion of the data.

The line inside the box represents the **median**.

The lines extending from the box are called **whiskers**.

Individual points outside the whiskers may represent **outliers**.

Applications

Box plots can compare:

- Marks of different classes.
- Salaries across departments.
- Test scores between groups.
- Age distributions.
- Product prices.

Advantages

Box plots are useful because they allow us to compare:

- Central tendency.
- Data spread.
- Variation.
- Distribution.
- Outliers.

Conclusion

Chart F contains multiple box-and-whisker structures representing different groups. Therefore, it matches Description 4.

Therefore: F $\rightarrow$ 4

G. Histogram

Matching Answer: G $\rightarrow$ 5

Description 5:

“Shows the distribution of a single continuous variable.”

Explanation

Chart G is a **Histogram**. A histogram is used to display the distribution of continuous numerical data.

The data is divided into intervals called **bins**. Each bar represents the number of observations that fall within a particular interval.

Characteristics of a Histogram

- Used for continuous numerical data.
- Data is divided into bins.
- Bars normally touch each other.

- Height represents frequency.
- Shows the shape of the distribution.

Examples

A histogram can show the distribution of:

- Student marks.
- Employee salaries.
- Customer ages.
- Heights.
- Weights.
- Examination scores.

What Can Be Identified?

A histogram can help identify:

- Most common values.
- Spread of data.
- Central concentration.
- Skewness.
- Possible unusual values.

Difference from Bar Chart

A histogram is used for **continuous numerical data**, whereas a bar chart is generally used to compare **separate categories**.

Conclusion

Chart G has adjacent bars forming a distribution pattern, which is characteristic of a histogram.

Therefore: $G \rightarrow 5$

H. Line Chart

Matching Answer: H → 7

Description 7:

“Shows how a value changes over time.”

Explanation

Chart H is a **Line Chart**. A line chart connects data points using lines to show changes in a variable over a sequence, especially over time.

The chart shown has a line that rises and falls at different points, indicating changes in the measured value.

Characteristics

- Uses connected points.
- Commonly represents time-series data.
- Shows increasing and decreasing trends.
- Makes changes over time easy to observe.
- Can be used with multiple lines for comparison.

Examples

Line charts can represent:

- Monthly sales.
- Annual profit.
- Daily temperature.
- Stock prices.
- Electricity consumption.
- Student performance over semesters.

Interpretation

If the line moves upward, the value is increasing.

If the line moves downward, the value is decreasing.

If the line remains almost flat, the value is relatively stable.

Conclusion

Chart H clearly shows a changing value through a connected line. Therefore, it matches Description 7.

Therefore: $H \rightarrow 7$

I. Stacked Bar Chart

Matching Answer: $I \rightarrow 1$

Description 1:

“Shows part-to-whole composition across categories.”

Explanation

Chart I is a **Stacked Bar Chart**. Each bar contains multiple colored sections stacked on top of each other.

The entire bar represents the total value, while each individual section represents a component of that total.

This makes the chart useful for comparing composition across several categories.

Example

Suppose a company wants to compare sales of three products across four quarters.

Each quarter can have one bar divided into:

- Product A.
- Product B.
- Product C.

The total bar represents total sales, while each section represents the contribution of a product.

Characteristics

- Contains multiple segments.
- Segments are stacked vertically.
- Each bar represents a category.
- Total height represents the whole.
- Individual segments represent components.

Applications

Stacked bar charts are useful for:

- Market share.
- Sales composition.
- Expense categories.
- Energy sources.
- Population composition.
- Budget allocation.

Conclusion

Chart I contains multiple stacked sections within each bar. Therefore, it represents part-to-whole composition across categories.

Therefore: I $\rightarrow$ 1

J. Area Chart

Matching Answer: J $\rightarrow$ 8

Description 8:

“Shows how a cumulative total changes over time.”

Explanation

Chart J is an **Area Chart**. An area chart is similar to a line chart, but the region underneath the line is filled.

The filled area emphasizes the magnitude or accumulated amount of a variable over time.

Characteristics

- Similar to a line chart.
- Area below the line is filled.
- Useful for showing trends.
- Can emphasize cumulative quantities.
- Commonly used with time-series data.

Examples

An area chart can show:

- Cumulative sales.
- Total energy generation.
- Revenue growth.
- Population growth.
- Website traffic.
- Electricity consumption.

Interpretation

If the filled area continuously increases, it indicates that the cumulative total is increasing over time.

In Chart J, the filled region increases as the horizontal axis progresses, indicating a cumulative growth pattern.

Conclusion

Therefore, Chart J matches Description 8 because the area chart shows how a cumulative total changes over time.

Therefore: J → 8

t displays changes over time, the **stacked bar chart** shows part-to-whole composition across categories, and the **area chart** emphasizes cumulative changes over time.

Thus, selecting the appropriate visualization helps present data clearly and allows important patterns, relationships, comparisons, and trends to be understood quickly.